

AUSTIN LIU

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EDUCATION

Brown University

B.S. in Applied Math-Computer Science, B.A. in Economics

Providence, RI

2027 / GPA 3.95

- **Relevant Coursework:** Computer Networks, Applied Cryptography, Operating Systems, Computer Systems, Machine Learning, Deep Learning, Data Structures and Algorithms, Statistical Inference

WORK EXPERIENCE

Persona Identities

Software Engineer Intern

May 2026 - August 2026

San Francisco, CA

- Incoming Summer 2026 Intern

Tato

Forward Deployed AI Engineer Intern

March 2026 - Present

Remote

- Generated **\$10,000+** in revenue within 3 months by architecting a full-stack rental module featuring customizable dashboards, integrated calendars, and tiered subscription logic; directly interfaced with clients to align features with business needs
- Improved dashboard responsiveness by **500ms** when rendering 10,000+ pet profiles by implementing **lazy loading** and efficient state management patterns within a high-performance **Next.js** and **Supabase** architecture
- Decreased feature development cycles by **30%** using **Claude Code** and compound engineering plugins to automate workflows

Symbotic

Software Engineer Intern

May 2025 - August 2025

Boston, MA

- Reduced **C#** API latency by **84%** by refactoring unbound queries into server-side execution, preventing CPU bottlenecks
- Ensured **99.95%** data integrity across **20M+** warehouse SKUs through automating health checks, preventing critical errors
- Cut SQL execution time for warehouse database by **60%**, converting full scans into index seeks using **SQL Sentry Explorer**
- Automated XML configuration management by deploying a **C# .NET** MVC application to **10+** warehouse sites nationwide, enabling seamless updates and reducing downtime by eliminating reliance on weekly versioned deployments

PROJECTS

IP/TCP | Go

April 2026 - Current

- Engineered **TCP stack from scratch**, implementing a sliding window protocol, 4-way handshakes, and packet reordering
- Designed a **RIP-based routing engine** utilizing the Bellman-Ford algorithm and split-horizon with poison reverse to achieve network convergence and prevent routing loops across multiple interfaces
- Optimized packet processing using a multi-threaded **Go-based network layer** with shared channels and mutexes

Guitar Tab Transformer | React, Django, OpenCV, TensorFlow, Python

June 2024 - Current

- Enhanced **dadaGP** open-source Python package by adding support for single-track input and irregular tunings, improving transcription accuracy across diverse guitar formats
- Improved sheet music processing efficiency by building an **OpenCV-TensorFlow pipeline** that applied morphological operations to clean and analyze **5K+** images
- Achieved **95% accuracy** in notehead classification by training **TensorFlow CNNs** for binary detection of noteheads

LEADERSHIP

Backend Software Lead

Brown Formula Racing

September 2024 - Current

Providence, RI

- Led a backend team of **10 engineers**, leading design discussions, task delegation, and mentorship to accelerate feature delivery
- Deployed a **React-Django** web application on **AWS EC2** with automated **CI/CD pipelines**, **Nginx** load balancing, and **AWS S3** storage, contributing to a **Top 20 national finish**

SKILLS

Languages: Python, SQL, Java, C/C++, C#, JavaScript/TypeScript, HTML/CSS

Technologies: React, Django, .NET, TensorFlow, OpenCV, Pandas, NumPy

Databases/Cloud: Firebase, Amazon Web Services, Google Cloud Platform, MySQL, Pinecone

Interests: Fingerstyle Guitar, Brown Men's Volleyball, Steak and Sushi, Sports